Caiyi Shen

Email: seleneshen123@outlook.com | Portfolio: https://seleneshen123.github.io/

Education

Northeastern University 09/2025 – 05/2027
M.S. in Game Science and Design *Boston, MA, United States*

University of Wisconsin-Madison 09/2021 – 08/2024
B.A. in Psychology; Certificate in Game Design *Madison, WI, United States*

Project & Research

Language Acquisition and Brain Laboratory (QLAB) | Research Assistant / Programmer 07/2026 – Present
Northeastern University *Boston, MA, United States*

BeforeHello - Communication Support for Autistic Adults (INSAR 2027 Annual Meeting Abstract)

- Collaborate with **Dr. Katherine Trice** to develop BeforeHello, an application that enables **autistic adults** to prepare and share personalized video introductions before first meetings in educational, healthcare, and other social settings, aiming to reduce first-impression bias.
- Collect and synthesize feedback from autistic users through think-aloud to identify usability concerns, accessibility needs, and opportunities for iterative improvement.
- Preparation and revision of a research abstract submitted to **INSAR 2027**, investigating how speaker neurotype and diagnostic disclosure influence novel word learning and retention.

Audiovisual Neural Entrainment - Language Learning Research

- Collaborate with **Prof. Zhenghan Qi** to develop a **PsychoPy**-based audiovisual language-learning experiment for **children**, drawing on the auditory statistical learning and neural entrainment paradigm of Batterink et al. (2023).
- Design low-stimulation, child-friendly animations and implement scheduled breaks and controlled pauses. Precisely synchronize artificial-language speech stimuli with visual animation cycles and integrate EEG event markers via Lab Streaming Layer.
- Conduct experimental timing validation, simulation testing, and iterative debugging. Develop detailed experimental protocols and troubleshooting documentation to support reliability and reproducibility.
- Conduct **EEG** experimental sessions from preparation through data collection, including equipment setup, electrode impedance checks, task administration, real-time signal monitoring using **mBrainTrain Streamer**, data recording and storage, and preliminary data inspection using **MATLAB**.

Time2Learn - Language and Literacy Research

- Facilitate research preparation and documentation for the Time2Learn study, including standardized assessments of language, literacy, and cognitive skills.

Boston Bilingual Pioneer Media | Project Lead / Game Designer / 3D Artist 03/2026 – Present
Northeastern University *Boston, MA, United States*

AI Learning Companion & Robotics | Chief Product Officer

- Lead product planning and software and hardware integration for AI-powered educational tools and learning companion robots.
- In charge of social media operation, product promotion, and market outreach across China and the United States.

Case-Based Psychological Assessment Learning

- Collaborate with **Prof. Chieh Li** to design an interactive **psychological assessment** course for doctoral students.
- Design game-based learning activities to support evidence interpretation, critical thinking, and assessment report writing.
- Analyze feedback from a 32-student course pilot (mean rating: 4.5/5) to inform instructional improvements.

Interactive Learning Platform

- Program and design interactive 3D courseware and game-based learning activities for the Axis AI platform.
- Conduct **quality assurance (QA)** testing for courseware, identify and troubleshoot issues in navigation, assessment workflows, and feedback systems. Implemented iterative software improvements.

Learning, Representations, & Technology Lab | Research Assistant 07/2022 – 09/2022
University of Wisconsin-Madison *Madison, WI, United States*

- Conducted think-aloud playtesting of a 3D astronomy learning game, documenting participants' navigation, verbal reasoning, and interpretation of scientific representations.
- Synthesized observations and transcripts into qualitative summaries of usability barriers, conceptual misunderstandings, and learning-design opportunities.

Psychology Clinic | Clinical Assistant 06/2023 – 08/2023
Peking University Sixth Hospital *Beijing, China*

- Observed and practiced administration of **transcranial magnetic stimulation (TMS)**, **repetitive TMS (rTMS)**, **theta-burst stimulation**, and **transcranial direct-current stimulation (tDCS)** under supervision.
- Assisted with **electroencephalogram (EEG)-based visual-stimulus experiments** to examine fear responses.
- Studied ethical principles and professional practices in clinical psychology, including informed consent, confidentiality, professional boundaries, and patient safety.
- Observed **gender-affirming mental health services**, with emphasis on culturally responsive communication, clinical assessment, and documentation related to hormone therapy.

UW Child Development Lab | Field Observer 09/2022 – 02/2023

University of Wisconsin-Madison *Madison, WI, United States*

- Conducted structured observations of preschool peer interaction and social communication behaviors through one-sided mirror, including verbal initiation, turn-taking, joint attention, toy sharing, cooperative play, and responses to peers' social bids.
- Connected classroom observations with developmental psychology literature in an analytical research essay synthesizing field observations with developmental psychology theories of preschool peer interaction, play-based learning, and early friendship formation.

Biodesign Challenge 2027 | Educational Game & Biomaterials Design 09/2026 – Present

Northeastern University *Boston, MA, United States*

- Lead the development of an educational tabletop game exploring chocolate production, child labor, and environmental sustainability, collaborating with a multidisciplinary team.
- Develop and test **mycelium and gelatin-based bioplastic** formulations to create physical game components.

Working Experience

Deeplight Games | Concept Artist & 2D Designer 09/2026 – Present

Generate @ Northeastern University *Boston, MA, United States*

- Develop concept art and 2D visual designs for a client-commissioned 3D action roguelike, translating client requirements and creative direction into cohesive visual concepts aligned with gameplay and production goals.
- Design dungeon environments, level layouts, and maps, incorporating environmental storytelling, player navigation, branching pathways, and combat encounters to enhance gameplay and exploration.
- Conceptualize characters, bosses, and abilities, integrating character aesthetics, gameplay mechanics, and environmental elements to establish distinctive and consistent visual identities.
- Contribute to UI/UX concepts and visual assets.

Bilingual Montessori School | Teacher 08/2024 – 08/2025

If Montessori School *Houston, TX, United States*

- Worked with preschool-aged children in a Bilingual Montessori classroom, observing attention, learning engagement, social interaction, and responses to hands-on learning materials.
- Used developmentally appropriate and Montessori-informed strategies to support children's autonomy, language development, self-directed exploration, and classroom participation.
- Developed a Chinese character learning game based on Montessori theory, observed patterns in children's visual attention, symbolic association, and interaction with learning materials.

Game Design and Analysis Class | Graduate Teaching Assistant 09/2026 – 11/2026

Northeastern University *Boston, MA, United States*

- Evaluate student coursework and provide rubric-based feedback on game mechanics, Mechanics–Dynamics–Aesthetics (MDA) analysis, player motivation, game balance, and prototype development.
- Guide students in connecting game design decisions to player experience and applying playtesting methods to evaluate and refine their prototypes.

Digital Media Content Creator 01/2024 – Present

NetEase Blizzard China Creator Partnership *Online*

- Produce gaming, cosplay, and livestream for online audiences, growing a social media account to 20,000+ followers.
- Create game-related videos through a creator partnership with **NetEase Blizzard China** and **Behavior Interactive**, with one video reaching 10,000,000 views.

Free Online Education Platform | Co-founder & Creative Director 08/2022 – Present

Schoolive Interactive *Online*

- Co-developed a free Moodle-based learning platform to share university-level educational resources with students in China, including high school students exploring college subjects.
- Developed and organized course content with attention to instructional sequencing, learner accessibility, and clarity of psychological and design-related concepts.

Skills

- Tools: R, R Studio, MATLAB, Python, HTML, Unity, Unreal Engine, Autodesk Maya, Blender, ZBrush, Substance Painter, QuadSpinner Gaea, Adobe Photoshop, Adobe Premiere Pro
- Languages: Mandarin (Native), English (Advanced), Japanese (Advanced, JLPT N2)

Awards & Accomplishments

- American Montessori Society (AMS)-Certified Teaching Assistant 2025
- Institute for Health Equity and Social Justice Research (IHESJR) Affiliated Graduate Student 2026
- PEAK Fellowship at Northeastern Univeristy: The Summit 2026
- Global Game Jam @ GDC Participant (1 of 100) 2027